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Exp-19: Implementation of Naive Bayes classification for Bank Loan prediction

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# Import required libraries
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
# Load sample dataset
data = load_breast_cancer()
X = data.data
y = data.target
# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create the Naive Bayes model
model = GaussianNB()
# Train the model
model.fit(X_train, y_train)
# Predict the test data
y_pred = model.predict(X_test)
# Display results
print("Actual Loan Status:")
print(y_test)
print("\nPredicted Loan Status:")
print(y_pred)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
```

print("\nAccuracy: {:.2f}%".format(accuracy * 100))

Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)

Output:

```
Actual Loan Status:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Predicted Loan Status:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 1 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Accuracy: 97.37%

Confusion Matrix:
[[40  3]
 [ 0 71]]
```